

RECON

Payment Reconciliation, Root-Cause Analysis & Automated Recovery

Presentation Explanation Report

Based on the RECON concept and architecture developed in the preceding discussion.

1. Executive Summary

RECON is an intelligent payment reconciliation and recovery API/service layer. It sits around a payment flow and follows a transaction from initiation through processing, capture, settlement and receiver confirmation.

If the intended receiver receives the payment successfully, the transaction is complete. If the transaction becomes inconsistent—for example, the payment is captured but the merchant is not credited—RECON correlates the events, identifies the likely upstream root cause, preserves evidence, initiates recovery such as a simulated refund, and creates an escalation case for the responsible bank/payment-operations department.

Core flow: Detect → Correlate → Diagnose → Recover → Escalate

2. The Problem

Payment systems are distributed across payment gateways, settlement services, webhook layers, receiver/merchant platforms and financial operations. A transaction can look successful in one system while failing in another.

In the demonstrated failure, a customer pays **■1,499**. The payment gateway reports **PAYMENT_CAPTURED**, but the settlement is routed to an unexpected account. The merchant receives **■0** and the order remains pending.

Expected receiver: ACC_MERCHANT_001

Actual settlement destination: ACC_TEMP_9931

Result: payment captured, merchant not credited.

3. Failure Lifecycle

Event	Meaning
PAYMENT_INITIATED	Payment initiation started.
PAYMENT_PROCESSING	Payment is being processed.
PAYMENT_CAPTURED	Payment was captured successfully.
SETTLEMENT_INITIATED	Settlement process started.
ACCOUNT_ROUTING_MISMATCH	Actual destination differs from expected receiver.
SETTLEMENT_COMPLETED	Settlement marked completed internally.
WEBHOOK_SENT	Webhook sent to expected receiver.
WEBHOOK_REJECTED	Receiver rejected webhook because funds were missing.
RECEIVER_NOT_CREDITED	Receiver confirmed funds were not received.

Event	Meaning
ORDER_PAYMENT_PENDING	Order remains stuck in pending state.
RECON_ANALYSIS_STARTED	Automated reconciliation begins.
ACCOUNT_ROUTING_MISMATCH_CONFIRMED	RECON confirms the root anomaly.
BANK_CASE_FORWARDED	Evidence-backed case is forwarded.
REFUND_INITIATED	Recovery/refund begins.
REFUND_COMPLETED	Refund completes in the simulation.

4. How RECON Solves It

- **Detect:** monitors transaction events and identifies inconsistencies.
- **Correlate:** connects related events instead of examining one error alone.
- **Diagnose:** determines the upstream event that best explains downstream failures.
- **Recover:** initiates a simulated refund when the merchant cannot correctly receive the payment.
- **Escalate:** creates a bank/payment-operations case containing transaction context and evidence.
- **Audit:** preserves the complete chronological transaction trail.

5. Root-Cause Analysis

RECON does not simply say 'payment failed'. It derives a causal chain from the event sequence:

PAYMENT_CAPTURED → ACCOUNT_ROUTING_MISMATCH → WEBHOOK_REJECTED → RECEIVER_NOT_CREDITED → ORDER_PAYMENT_PENDING

The routing mismatch is treated as the upstream root cause, while webhook rejection, missing receiver credit and the pending order are downstream consequences.

6. Automated Recovery and Escalation

After confirming the failure, RECON performs two recovery actions in the simulation:

- **Refund:** return ■1,499 to the payer and update the transaction to **REFUNDED**.
- **Escalation:** create a bank/payment-operations case containing the transaction, expected receiver, actual destination, root cause and evidence trail.

This changes the workflow from a passive alert into an actionable recovery process.

7. RECON as an API / Service Layer

The best product description is an **API/service layer around payment infrastructure**. A merchant or payment application can expose transaction events to RECON; RECON monitors the transaction until it reaches a valid terminal state.

Input	RECON	Output
Transaction + events	Monitor / correlate	Unified transaction state
Settlement data	Detect mismatch	Root-cause signal
Receiver/webhook results	Analyze sequence	Reconciliation result
Failed reconciliation	Trigger recovery	Refund status
Failure evidence	Create escalation	Bank/payment-ops case

8. Merchant Dashboard

RECON MERCHANT is the investigation dashboard. It shows payment status, expected amount, settlement mismatch, transaction metadata and the chronological event timeline.

After reconciliation it can add a **Refund & Recovery** section showing **REFUND COMPLETED**, the refund amount, masked identifiers, **BANK CASE FORWARDED**, the root cause and the recovery action.

The backend remains the source of truth. The frontend only displays the state returned by the backend.

9. Payer Experience

RECON PAY represents the customer-facing flow. The customer initiates a ■1,499 payment. If the backend determines that the payment cannot correctly reach the merchant, the recovery state is reflected by the backend. The refund decision is not made by the frontend.

10. Sensitive Data Masking

The dashboard can mask sensitive identifiers by default, for example **TXN-****-6DA5** and **ACC_****_001**, with an authorized Reveal action. The backend/API still uses the full transaction identifier for lookup and correlation.

This demonstrates observability while reducing unnecessary exposure of sensitive identifiers on screen and in logs.

11. Architecture

Merchant / App → RECON API → Payment & Settlement Events → RECON Engine → Detection → Analysis → Recovery / Escalation → Audit Trail → Merchant Dashboard

The interfaces visualize the system, while the backend reconciliation engine is responsible for deciding whether a transaction is recovered, escalated or refunded.

12. Hackathon Demo

- Customer initiates a ■1,499 payment.
- Payment is captured successfully.
- Settlement is deliberately routed to the wrong simulated account.
- Merchant dashboard shows ■0 received and the routing mismatch.
- Webhook and receiver failures appear in the timeline.
- RECON correlates the event chain and confirms the root cause.
- A simulated bank case is created/forwarded.
- Refund is initiated and completed.
- The transaction reaches **REFUNDED** and the merchant sees the recovery trail.

13. 30-Second Judge Explanation

"RECON is an automated payment reconciliation and recovery layer. Imagine a customer pays ■1,499. The payment gateway successfully captures the payment, but a settlement-routing error sends the funds to the wrong destination, so the merchant receives nothing. RECON monitors the complete event chain, identifies the routing mismatch as the root cause, creates an evidence-backed case for the responsible bank or payment-operations team, and automatically initiates recovery by refunding the payer. The merchant gets one unified timeline showing what

happened, why it happened, and how the system recovered."

14. What RECON Is — and Is Not

- **It is:** a payment reconciliation, root-cause analysis, monitoring and recovery service/API concept.
- **It is:** a unified transaction-correlation and audit layer.
- **It is:** an automated recovery and escalation workflow.
- **It is not:** a replacement for a real bank, payment gateway or settlement network.
- **Hackathon implementation:** payment, settlement failure, bank escalation and refund are simulated to demonstrate the architecture and recovery logic.

15. Final Product Statement

RECON turns a silent or ambiguous payment failure into an actionable recovery workflow.

It follows the payment, finds the inconsistency, identifies the likely root cause, preserves evidence, escalates the issue to the responsible financial operations team, and recovers the payer's money when the merchant cannot be correctly credited.